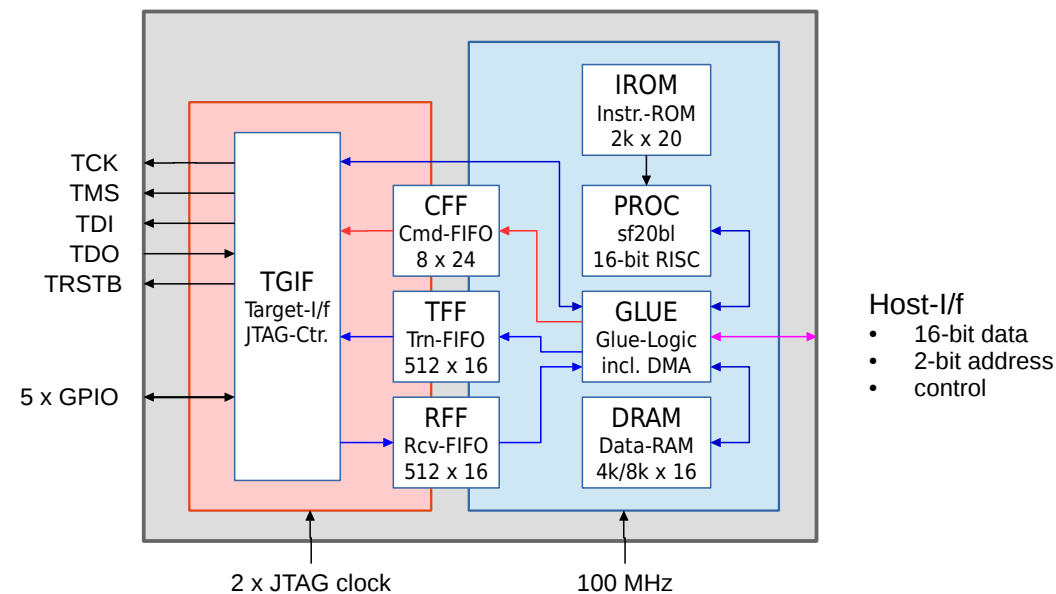


Concept /Proposal

Bscan: MSP2100/MSP3000 channel controller for high-speed boundary scan



Concept

- Hardware JTAG engine (TGIF) for IR/DR shifts and TAP state sequences (TMS sequences)
 - Data In/Out from 1kiB FIFOs → up to 4096-bit shifts with queuing and seamless JTAG operation
 - Independent Host-I/f clock and JTAG clock, up to 100MHz JTAG clock (up to 200MHz TGIF clock)
- Local Processor to handle JTAG data flow and boundary scan operations
 - Writes TDI data to transmit FIFO and reads TDO data from receive FIFO (DMA alternative possible)
 - Checks received TDO data, e.g. SVF style mask/value compare
 - 100MHz fixed clock
- Host Interface with DMA to the data RAM (DRAM) of the local processor
 - Read/write of scan vectors, masks and expected results
 - Read/Write to status/command queue (bootloader communication concept)
- MSP2100 FPGA configurations
 - DualBscan: 2k words Instr.-ROM, 16kiB Data-RAM
 - QuadBscan: 2k word Instr.-ROM, 8kiB Data-RAM, feasibility (logic size) to be checked

Performance / Timing example

- Example: 2048-bit scan chain → 128 x 16-bit words
 - Write TDI data to TFF: $2 * 128 = 256$ cycles
 - Read TDO data from RFF: $2 * 128 = 256$ cycles
 - Check TDO data (mask/value) with OK/NOK result: $6 * 128 = 768$ cycles
 - Command interpreting, status update, loop-start overhead: 300 cycles (estimated)
 - Total for one boundary scan: ~1600 cycles
- At the max JTAG speed of 100 MHz the scan takes 2048 cycles + some overhead
 - → seamless scan operations including result checking by local processor is possible
 - Prerequisite is that the Host MCU provides input, mask and expected value data at the target scan rate, e.g. for 100 MHz JTAG clock $300 \text{ Mbits/s} = 37.5 \text{ Mbytes/s}$ → for the MSP2100 the data transfer will be a bottleneck: 2 channels with max. 25MHz or one channel up to 50 MHz